Hadassah Hospital in Jerusalem, doctors have pioneered a novel treatment for patients suffering from acute depression. They implant electrodes into the patient's brain, and wire the electrodes to a minuscule computer implanted into the patient's breast. On receiving a command from the computer, the electrodes use weak electric currents to paralyse the brain area responsible for the depression. The treatment does not always succeed, but in some cases patients reported that the feeling of dark emptiness that tormented them throughout their lives disappeared as if by magic.

One patient complained that several months after the operation, he had a relapse, and was overcome by severe depression. Upon inspection, the doctors found the source of the problem: the computer's battery had run out of power. Once they changed the battery, the depression quickly melted away.[5]

Due to obvious ethical restrictions, researchers implant electrodes into human brains only under special circumstances. Hence most relevant experiments on humans are conducted using non-intrusive helmet-like devices (technically known as 'transcranial direct current stimulators'). The helmet is fitted with electrodes that attach to the scalp from outside. It produces weak electromagnetic fields and directs them towards specific brain areas, thereby stimulating or inhibiting select brain activities.

The American military experiments with such helmets in the hope of sharpening the focus and enhancing the performance of soldiers both in training sessions and on the battlefield. The main experiments are conducted in the Human Effectiveness Directorate, which is located in an Ohio air force base. Though the results are far from conclusive, and though the hype around transcranial stimulators currently runs far ahead of actual achievements, several studies have indicated that the method may indeed enhance the cognitive abilities of drone operators, air-traffic controllers, snipers and other personnel whose duties require them to remain highly attentive for extended periods.[6]

Sally Adee, a journalist for the *New Scientist*, was allowed to visit a training facility for snipers and test the effects herself. At first, she entered a battlefield simulator without wearing the transcranial helmet. Sally describes how fear swept over her as she